
CBE 30355 Transport I Final Exam

December 12, 2025

Closed Books and Notes

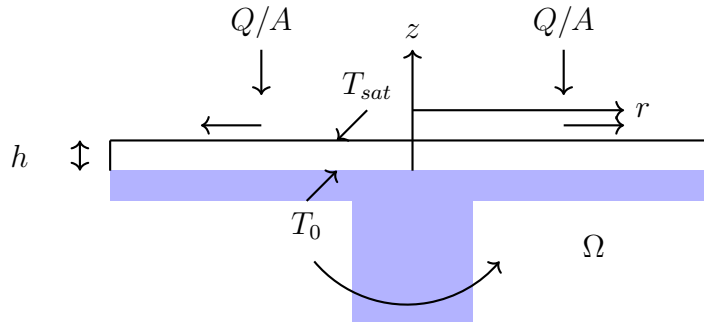
Problem 1

(20 points)

Surface Deposition and Film Depth Control

Cold rotating disks are used in vacuum systems to condense and trap residual vapor. Rotation prevents excessive condensate buildup by centrifugally draining the film. In this problem you are asked to determine the equilibrium film depth as a function of the parameters of the problem. Suppose fluid with kinematic viscosity ν is condensing on a disk of radius R with a uniform volumetric rate Q/A (e.g., units of velocity). The disk is rotated with an angular velocity Ω . Balancing deposition with the centrifugal flow, determine the following:

- a. For thin fluid layers the rotational velocity is just $u_\theta = \Omega r$ over the entire film (i.e., it moves with the rotational velocity of the disk). Using this, determine the differential equations and boundary conditions which govern the radial and vertical velocity distribution in the film.
- b. Explicitly solve these equations for the equilibrium film depth as a function of the deposition rate Q/A . (Hint: Think about how Q/A relates to the vertical velocity at $z = h$!)
- c. In your analysis you have assumed that the film is thin enough that the θ velocity is uniform. This is usually pretty good, but it does impose an upper limit on the rotation rate. What is the maximum rotation rate for a given film depth h for which this would be valid?
- d. In order for the vapor to condense into a liquid at the surface of the film, it must give up its latent heat of vaporization. Under some conditions this is simply due to thermal conduction through the film of thickness h . If the surface temperature of the film is T_{sat} , the disk temperature is T_0 , the thermal conductivity of the liquid film is k and the latent heat of vaporization of the liquid per unit volume is ρL (units of energy/volume here), "close the loop" to determine Q/A and the equilibrium film depth h . Ignore any convection in calculating the heat flux - conduction only! Think about the direct analogy between conductive heat transfer and the momentum flux in simple shear flow you have solved many times!



The following equations may be useful: just remember that the vast majority of the terms are either negligible or zero!

$$\rho \left(\frac{\partial u_r}{\partial t} + u_r \frac{\partial u_r}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta^2}{r} + u_z \frac{\partial u_r}{\partial z} \right) = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial(r u_r)}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_r}{\partial \theta^2} - \frac{2}{r^2} \frac{\partial u_\theta}{\partial \theta} + \frac{\partial^2 u_r}{\partial z^2} \right] + \rho g_r$$

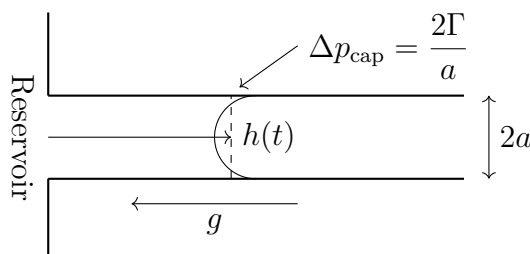
$$\rho \left(\frac{\partial u_\theta}{\partial t} + u_r \frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r u_\theta}{r} + u_z \frac{\partial u_\theta}{\partial z} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial (r u_\theta)}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_\theta}{\partial \theta^2} + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} + \frac{\partial^2 u_\theta}{\partial z^2} \right] + \rho g_\theta$$

$$\begin{aligned} \rho \left(\frac{\partial u_z}{\partial t} + u_r \frac{\partial u_z}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_z}{\partial \theta} + u_z \frac{\partial u_z}{\partial z} \right) &= -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2} \right] + \rho g_z \\ \frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r \rho u_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho u_\theta) + \frac{\partial}{\partial z} (\rho u_z) &= 0 \end{aligned}$$

Problem 2 (20 points)

Capillary Filling of a Tube by Surface Tension Against Gravity

Consider the **vertical** circular tube of radius a depicted below (note that the image is rotated so that it fits on the page!). At time $t = 0$ the end of the tube is dipped into a liquid reservoir (viscosity μ and density ρ). If the fluid wets the tube then capillary forces draw the fluid into the tube, where the pressure differential at the meniscus is given by $\Delta p_{cap} = 2\Gamma/a$ where Γ is the surface tension. Eventually, gravity brings the process to a stop. Here we analyze this problem.



- Write down the governing equations and boundary conditions, as well as the equation governing the time dependent filled length h .
- Determine the equilibrium height of fluid in the capillary at long times (this is easy!).
- Render the equations dimensionless to determine how the filling time depends on the parameters of the problem.
- Solve the problem to determine how long it takes for the fluid to get 90% of the way to equilibrium. You may leave your final result in integral form if you wish, although it is fairly simple to get t^* as a function of h^* .

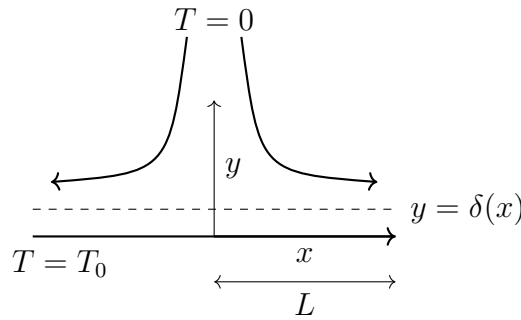
(Hint: Inertial effects are negligible for this problem, and the pressure differential is applied only over the filled length h of the capillary, which changes in time at a rate equal to the average axial velocity!)

Problem 3

(20 points)

Impinging Planar Jet and Thermal Boundary Layer

Consider the heat transfer from a plate in the flow due to an impinging planar jet as depicted below. The fluid from the jet (usually produced by flow from a slit) has a temperature $T = 0$ (e.g., we subtract it off as a reference) and impinges on a plate maintained at a temperature T_0 . Your mission is to determine the total heat transfer from the plate per unit extension into the paper, Q/W .



$$\rho C_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

$$\frac{Q}{W} = 2 \int_0^L -k \left. \frac{\partial T}{\partial y} \right|_{y=0} dx$$

- The velocity in the x -direction near the plate is given by $u = \lambda xy$. Applying no-slip and continuity, what is the velocity v in the y -direction?
- The heat transfer equations are given above. Scale the equations in the boundary layer limit to determine the boundary layer thickness δ and how it depends on x (or L). Scale the expression for the integrated heat flux to determine the total rate of heat loss from the plate to within some unknown $\mathcal{O}(1)$ constant.

Problem 4

(20 points)

Irrigation Pumping with CP-80i: Flow Rate and Efficiency Estimation

You are tasked with using the pump CP-80i (pump curve below) to pump water 10 m uphill to fill a reservoir for irrigation purposes. The problem is that you only have a 10 kW power supply for the pump and need to determine what flow rate you are going to get for the system. So:

- a. If you have a really large diameter pipe for your system, pipe losses will be negligible. In this case, what flow rate can you expect (really the upper limit) and what would the pump efficiency be?
- b. The proposed piping network consists of 40 m of 3" schedule 40 PVC tubing (ID = 7.72 cm), together with 6 90° elbows and a safety grate at the inlet with $K = 1.5$. Given this, get a better estimate of the flow rate you can expect from the pump (still running at 10 kW). Note that this would require a few iterations to get the exact value, so just show your work and try to get close in a short amount of time. Kick the iteration off by guessing that losses are about equal to the change in height (not a bad guess, actually) and using the appropriate flow rate from the pump curve. I'm looking for method here!
- c. The overall efficiency of the system would be the useful work out (e.g., the potential energy gained by the water by pumping uphill) divided by the energy into the pump. Note that this is intrinsically lower than the pump efficiency. What is this overall efficiency based on the flow rate you calculate in part b?
- d. The piping leading to the pump from the source consists of two elbows, the safety grate, and 4 meters of pipe. Based on this, what recommendations can you make about the placement of the pump relative to the water source?

Useful relations:

$$h_L = \left(\sum K + \frac{4f_f L}{D} \right) \frac{u^2}{2g}$$
$$f_f = \begin{cases} \frac{16}{Re} & Re < 2100 \text{ (laminar)} \\ \frac{0.0791}{Re^{1/4}} & 3000 < Re < 10^5 \\ \frac{1}{\sqrt{f_f}} = 4.0 \log_{10}(Re \sqrt{f_f}) - 0.4 & Re > 3000 \end{cases}$$

Fitting K values:

- Sudden expansion: 1.0
- 90° elbow: 0.90
- Safety grate: 1.5

